DEVELOPMENT OF AN AGENTIC AI SYSTEM FOR PROJECT PLANNING AND MANAGEMENT IN A DIGITAL COLLABORATIVE ENVIRONMENT

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ABSTRACT

Project management and scheduling are critical challenges in construction and civil engineering, where complex demands in coordination, information management, and task prioritization frequently lead to delays, obstacles, and budget overruns. Recent advances in agentic artificial intelligence (AI) systems demonstrate significant potential for handling complex decision-making in large-scale collaborative environments. In this pilot study, we developed and evaluated a dual-agent AI system capable of real-time planning and management of human teams in a Minecraft-based structure-building evaluation environment. A three-person team was evaluated over nine rounds, tasked with replicating template structures in the least amount of time. For each round, one of three management conditions was used: AI project manager, human project manager, and no project manager oversight. Results indicate that while the No PM condition achieved the fastest overall completion times ($431 \pm 90s$), the AI PM demonstrated the shortest planning phase ($93 \pm 22s$) and maintained consistent plan quality scores (4 ± 0). These findings provide initial empirical evidence for AI systems' potential in project management roles and offer insights into optimal human-AI collaboration dynamics in digital collaborative environments. We release the complete agentic framework and Minecraft evaluation environment as open-source software to facilitate future research in AI-assisted project management.

INTRODUCTION

In recent years multimodal foundation models, AI models trained for general use on a large-scale data sources, have made extremely capable models accessible to a wider audience. Both through chat interfaces and API-based interactions, these models are becoming essential productivity tools, spurring the growth of agentic applications. Agentic use of models typically involves an AI performing a specific, but broad, task which would generally be difficult for a human to implement programmatically. Agentic applications have seen a surge of applications in coding (Novikov et al. 2025), medicine (Gao et al. 2024), and novel research (Schmidgall et al. 2025). These agentic applications are driven by LLM and Multimodal models' ability at complex problem solving and analysis given a large data context input, often outperforming humans in these tasks (OpenAI 2024).

Recognizing the potential for these agentic systems, we aimed to evaluate the performance of such systems in the domain of construction project planning and human management. To simplify the evaluation environment we turned to the very popular block-based sandbox game Minecraft (Mojang Studios 2011), where players have the ability to creatively build and design structures from a variety of in-game blocks which can also be processed and modified to meet their architectural desires. To simulate a construction project, we designed a multi-round mini-game inspired by the MC Championship mini-game series "Big Sales at Build Mart" (MC Championship Fandom Contributors 2025) (MC Championship Fandom Contributors 2024) where players must exactly replicate a provided template structure in the shortest amount of time. In each round, the players must gather the necessary resources and process those resources to replicate the template structure. Thus the designed mini-game aims to provide an effective evaluation environment for construction planning capabilities where optimal planning and personal management is rewarded with a shorter completion time.

Alongside this evaluation environment, we also developed an agentic framework, which we refer to as the AI PM in this paper, using the OpenAI Agents SDK (OpenAI 2024) to collect relevant information about the structure, the map locations, and player details in order to generate an optimal plan for the players to follow. Within our framework we use two agents, a high-level planning agent which devises an optimal plan for completing the structure in minimal time and a low-level manager agent which provides a task-level player instructions given the generated plan. We include human-managed rounds and unmanaged rounds as a baseline comparison for evaluating the AI PM's performance.

The latest agentic framework, Minecraft files, and additional setup instruction can be found on Github at: `https://github.com/AIConLab/ai-pm`. This codebase is still under active development as we prepare for larger user-studies.

In summary our contributions are as follows:

- Developed an agentic AI system for plan generation and dissemination based on real-time environment data and contextual building design data.
- Developed and released a novel planning and management evaluation environment in Minecraft.

- Conducted a pilot study to gauge effectiveness of AI planning and management for digital collaboration tasks when compared to human-managed cases and no-management cases.

RELATED WORK

AI in Planning and Management

The use of AI for planning and management has seen broad applications and techniques deployed in recent years. We direct the reader to (Adamantiadou and Tsironis 2025) for a recent survey of the field. For example, we see works such as (Agarwal et al. 2011) deploying a genetic algorithm and neural network for solving resource-constrained project scheduling problems. In construction projects specifically, approaches such as fuzzy discrete event simulation (Sadeghi et al. 2016), Machine learning based approaches to improving the performance of the Probabilistic Critical Path Method (PCPM), and artificial neural networks (Jha and Chockalingam 2011) have been used to enhance project planning and monitoring. Language foundation models like GPT-4 have also seen use in planning applications. In (Liu et al. 2023) the authors deploy GPT-4 in conjunction with the Planning Domain Definition Language (PDDL) in order to solve classical planning problems. The potential for AI-based management extends beyond planning however, we also see its capabilities in real-time communication management of multi-person information flows (Rosenberg et al. 2025).

AI in Digital Environments

Digital environments like video games serve as the perfect test bed given their fixed rule sets and predictable difficulty scaling (Laird and VanLent 2001). Early works in AI for in digital environment planning such as, (Kovarsky and Buro 2006) and (Churchill and Buro 2011), describe the use of AI for optimal automated build-order planning in real-time strategy games like Starcraft. Minecraft has become a popular testbed for more recent AI research due to it's open-ended nature and large action space for creative decision making. Various works use Minecraft as an evaluation platform for AI agent's long-term learning capabilities, testing the agent's ability to progress in the game environment autonomously (Wang et al. 2023) (Wang et al. 2024).

METHODOLOGY

Study Design

As mentioned in the introduction, the proposed evaluation environment was inspired by the MC Championship mini-game "Big Sales at Build Mart" (MC Championship Fandom Contributors 2025). Our study makes use of nine rounds with a single structure to replicate per round. Figure 1 provides a summary of each round's structure and management style.

The game map, visualized in Figure 2 includes the following:

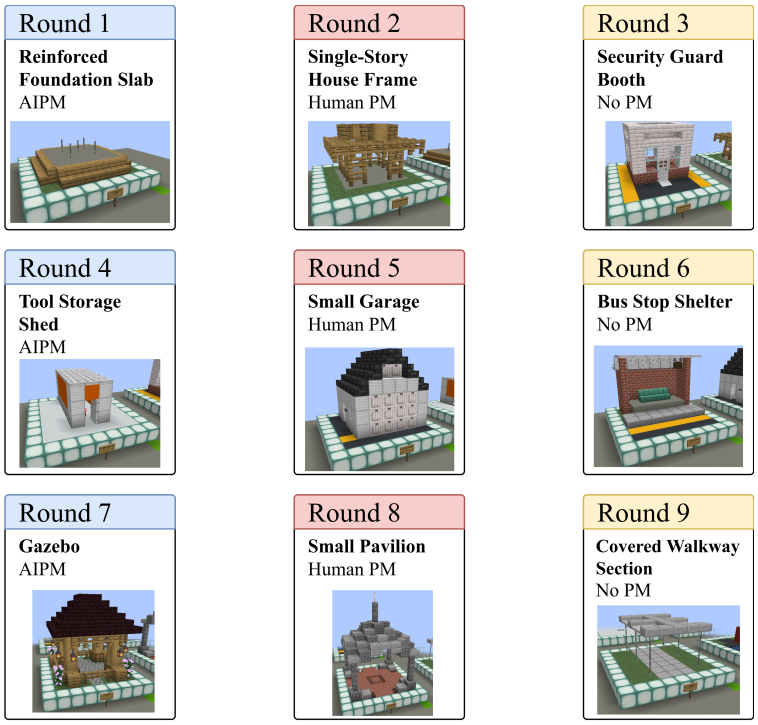


Figure 1. A summary of the nine rounds. Includes the management style for each round and an image of the structure to replicate.

- Game Lobby: Where players spawn upon joining the server and which they are teleported to after completing a round.
- Round Map: A bounded area which contains the Processing area, Structure Template, Build Area, and the Resource Area.
- Structure Template: Inside the Round Map. The completed structure which the players use as reference for their build replication.
- Build Area: Inside the Round Map. The area where players must replicate the Structure Template.
- Resource Area: Inside the Round Map. The area containing all necessary blocks to replicate all designed structures.
- Processing Area: Inside the Round map. The area where users are provided PPE, tools, and materials for processing blocks to complete the round.

A single team of three human players must collaborate to exactly replicate a template structure each round. In this pilot study, researchers from the author's lab serve as the human participants. A tutorial structure is completed before the evaluation begins to allow for all players to become familiar with the map layout and game rules. Round transitions are managed by an administrator in-game who toggles the logic circuit for starting a particular round. Game logic is managed through command blocks, blocks in Minecraft which allow for commands to be run, which include

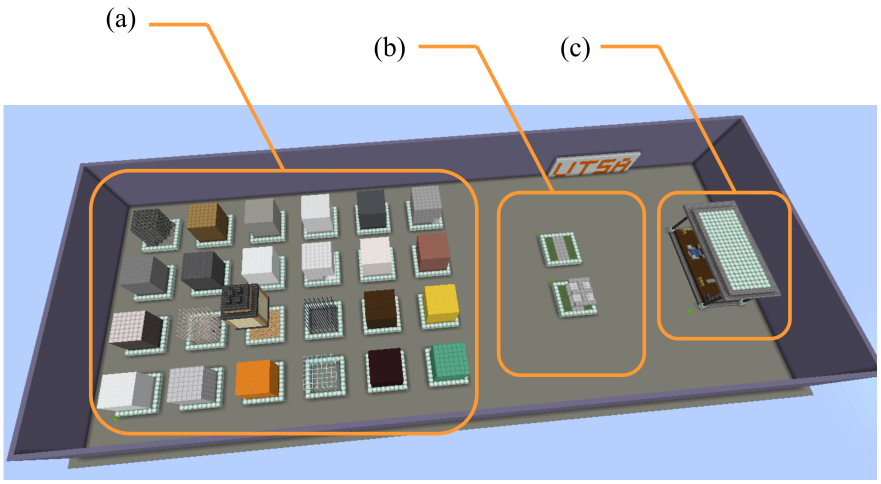


Figure 2. The Round Map: (a) Resource area. (b) Structure Template and Build Area. (c) Processing Area.

custom built functions from a Data pack, a Minecraft file system allowing for external function files to be executed in-game.

For normal rounds, the players are teleported to the Round Map and a timer begins. AI PM rounds trigger the agentic system to generate a plan given the current game context, this is described in detail in the following section. In the Human PM rounds, a fourth player who cannot build or interact with the world begins to announce a plan generated before the game for their designated rounds. In the No PM rounds, the players must collaborate to generate plans as they start the round. Collaboration by participants is done entirely through text via Minecraft's in-game chat.

We gather user data in the form of three surveys throughout our study. These surveys are shared with participants through the in-game chat as hyperlinks. Before the game begins, the players must complete a pre-game survey to assess their prior experience in collaborative digital environments. We survey the players after each round to assess their perceived difficulty of the completed round and the effectiveness of the planning and management provided that round. Finally, a survey is provided post-game to evaluate the players perception of cooperation, game effort level, and general feedback of the game.

AI PM System Design

Figure 3 provides a high-level system diagram of the developed AI PM system. The developed AI PM system makes use of two agents, a planning agent and a management agent. A state of the art (SOTA) reasoning model, GPT-5, is selected as the planning agent since reasoning models have shown the ability to successfully plan across large, ambiguous, problem spaces (Li et al. 2025). The management agent in our current framework serves to breakdown the high-level plan generated by the planning agent. We use GPT-4.1-mini as the manager agent model as it balances intelligent instruction following and generation speed.

For real-time Minecraft data extraction, we use an event-based program and a polling program. The event-based program reads from a log file which Minecraft auto-generates and updates internally.

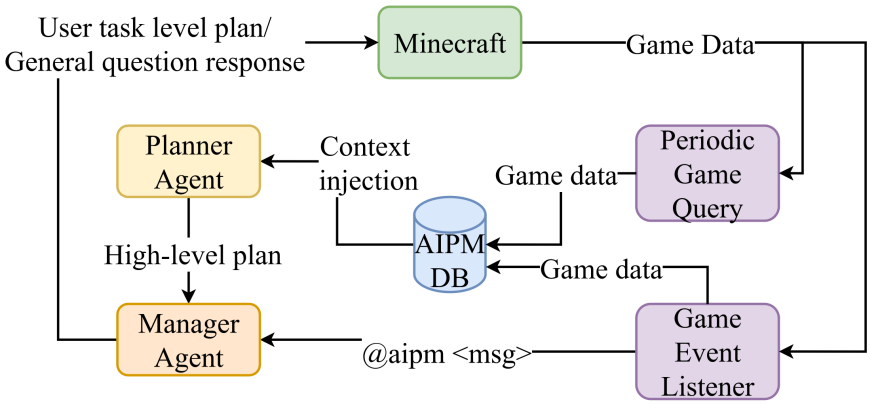


Figure 3. System diagram for the AI PM system.

From the log file the event-based program can handle players joining and leaving the server, as well as read any in-game chat messages. The polling program queries the game state every 3 seconds and checks for player data including inventory and location, and round state information like the current active round number.

The data is filtered to either generate an AI PM response if the user asked a question or is written to a SQLite database. From the in-game chat, a player can type `@aipm <msg>`, in order to ask the AI PM any Minecraft related question such as how to craft a certain item, etc. The database is initially populated with game context data such as structure recipes (the blocks required to replicate a structure) and XYZ locations of key map items; the database is also continuously updated with live round data during game play.

Once an AI PM round is active, current round and player data is injected into a planning prompt and a request is made using the OpenAI Agents SDK (OpenAI 2024). The generated response represents a high-level plan to optimally replicate the round's template structure in the minimum amount of time; the high-level plan, however, does not provide user-level instructions for how to proceed – for that the manager agent is required. The manager agent's prompt is injected with the latest high-level plan, round data, and player data and a response is generated which breaks down the high-level plan into specific instructions for each player. An example of these plans is visualized in Figure 4. The instructions are announced in-game via Minecraft's text-based chat. This process is repeated from scratch for each round where the AI PM is assigned.

RESULTS

Study Statistics

The round completion times for the pilot study are depicted in Figure 5a. We split the round completion time between the time taken for planning and execution, where planning time is the time elapsed until a plan has been communicated in the Minecraft chat and execution time is the remaining time until the round has been completed. For the No PM case we manually review the study footage to determine when a plan has been created between the study participants in the in-game chat. The post-round survey responses are visualized in Figure 5b. We determine the plan

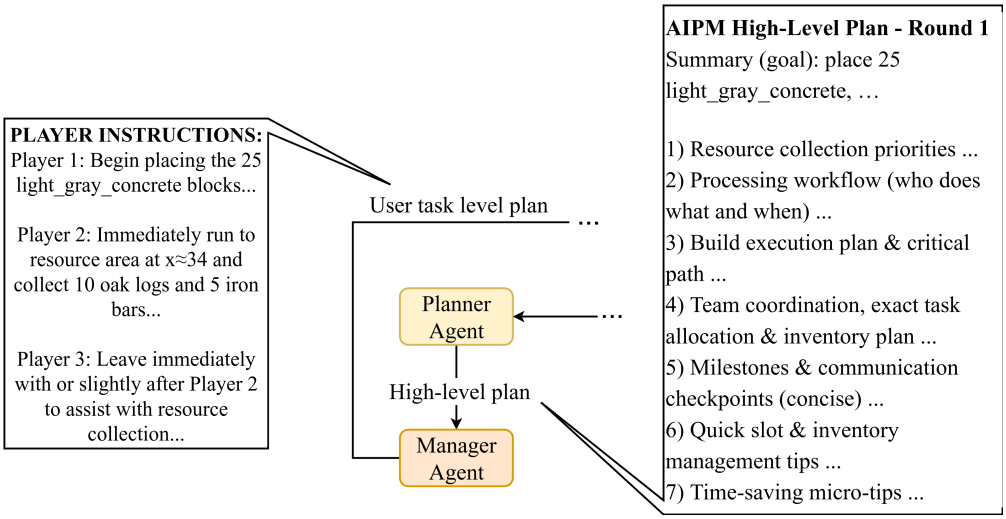
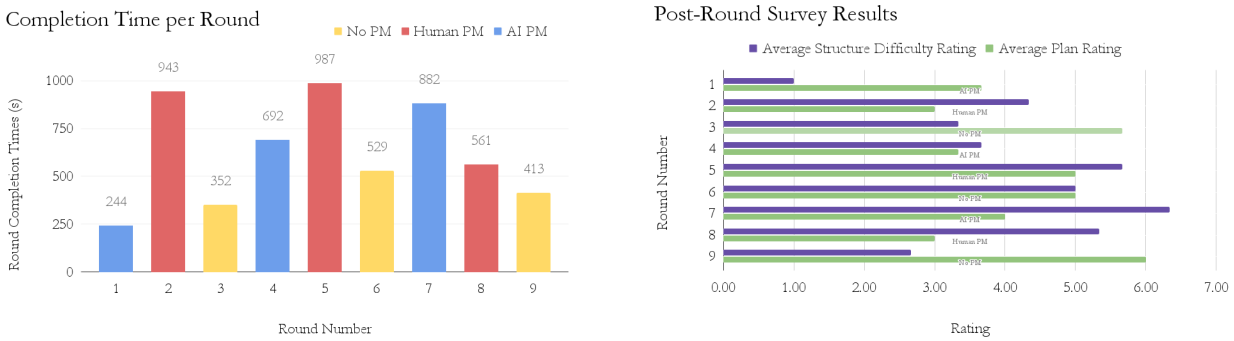


Figure 4. Example snippets of AI generated plans.

rating for each round by averaging the all responses for a round. Plan quality by management style is determined by then averaging the plan ratings for each round class.



(a) Total completion time in seconds for each round.

(b) Averaged responses from the post-round survey.

Figure 5. Round performance metrics showing completion times and participant ratings. Plan quality and structure difficulty are rated on a 1-7 scale where 1 is worst quality/least difficult and 7 is best quality/most difficult.

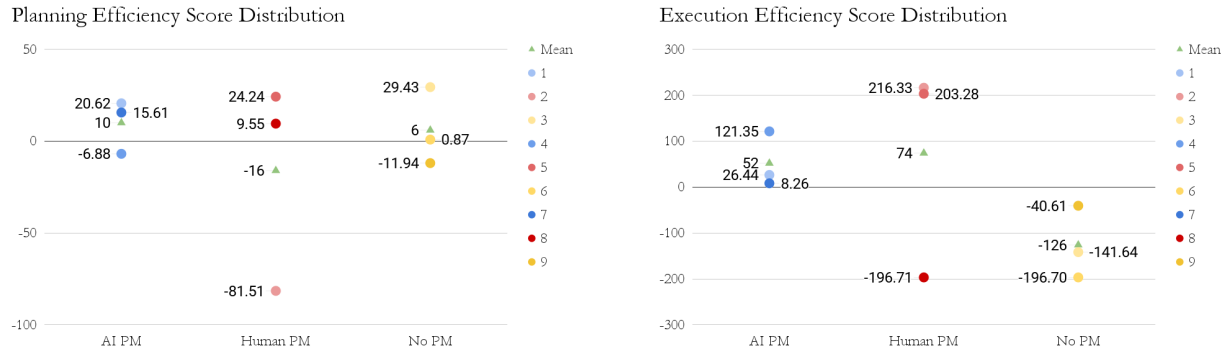
To compare the efficiency of each round independent of the round difficulty, we use a residualization technique described in (García et al. 2024). A linear regression is applied to model the expected time given the round difficulty, the residual of the time variable for each round is then calculated to be interpreted as an adjusted score not explained by the round difficulty – we call this the Efficiency Score for a round. We multiply the score by -1 for better interpretability such that a higher value indicates the round completed faster than expected and lower score indicates the round completed slower than expected. The complete formula for the Efficiency Score is:

$$\text{Efficiency Score}_i = ((\beta_1 x_i) + \beta_0) - y_i \tag{4.1}$$

Where for each round i:

- y_i is the observed time.
- x_i is the difficulty rating.
- β_1 is the slope and β_0 is the intercept of the regression line, representing the overall relationship between time and difficulty.

We calculate an efficiency score for both planning times and execution times. These are visualized in Figure 6a and Figure 6b respectively. The key study statistics are summarized in Table 1.



(a) Planning efficiency distribution.

(b) Execution efficiency distribution.

Figure 6. Efficiency score distributions for each round. Values represent the calculated efficiency score, where a higher value is better. The mean score is also plotted for each condition.

Table 1. Performance summary statistics by management condition. Values presented as mean $\pm$ standard deviation. Bold indicates top performer for column. Efficiency scores represent difficulty-adjusted performance (higher = better efficiency). Plan Quality rated on 1-7 scale (higher = better quality).

Condition	Total Time (s)	Planning Time (s)	Execution Time (s)	Structure Difficulty	Planning Efficiency	Execution Efficiency	Plan Quality
AI PM	606 ± 328	$\mathbf{93 \pm 22}$	513 ± 309	4 ± 3	$\mathbf{10 \pm 15}$	52 ± 61	4 ± 0
Human PM	830 ± 234	126 ± 54	704 ± 224	5 ± 1	-16 ± 57	$\mathbf{74 \pm 235}$	4 ± 1
No PM	$\mathbf{431 \pm 90}$	97 ± 22	$\mathbf{334 \pm 75}$	4 ± 1	6 ± 21	-126 ± 79	$\mathbf{6 \pm 1}$

Analysis

In terms of total completion time the No PM case proved to provide the best performance. Interestingly, the No PM case was not the best performer in either planning or execution efficiency. We believe this may be because the No PM rounds did not have a significantly difficult round based on the rating distribution in Figure 5b, though additional statistical tests would be needed to validate this. In addition, participants rated the plans from the No PM cases to be of the best quality although the efficiency scores do not reflect this to be the case. This may be due to user bias as plans in those cases came from internal cooperation between the players; less verbose plans could also affect this rating as AI PM and Human PM plans were generally more thorough and required more reading, a fact indicated by a majority of the post-study survey responses. Another potential factor for the faster No PM times could be that of user experience. Two of the participants rated themselves as very experienced with Minecraft (6 out of 7 in pre-study survey), while the third participant rated themselves as not very experienced with the game.

Overall the AI PM case did show promising potential as it performed the second best in terms of overall completion times and second best for execution efficiency, for planning efficiency it achieved the highest score. Study participants rated the plan quality of both the AI PM and Human PM equally on average. The post-study survey indicates participants desired less verbose plans and better task-level breakdown of plans, but user's appreciated the precise block quantities and processing steps specified in the AI PM plans. Although the AI PM plans do include user-level instructions (specifying the user's name and exactly what they should do) our current method for transmitting the AI generated text to the Minecraft chat could be contributing to the lower perceived plan quality. Future work could perform better formatting syntax and expression separation for improved readability in-game.

The Human PM case achieved the slowest overall completion times, but did perform the best in terms of execution efficiency. This suggests that the human-led team, while slower overall, performed better than expected on relative to the task difficulty. This could potentially be due to better generated plans, although the users perceived the plan quality to be about the same on average as the AI PM plans. The post-study survey indicated that participants appreciated the concise and direct format of the Human PM plans but felt it did not provide specific block quantities as well as other details.

CONCLUSION

In this paper we presented a novel evaluation environment for planning and management capabilities which allows for AI planning and management capabilities to be compared to human performance. An agentic management system is developed to allow for real-time communication with the in-game participants. A pilot study was then conducted which showed promising results for the AI PM's management and planning performance, with it resulting in the most efficient planning method and outperforming a Human PM in terms of total completion times.

Limitations

One of the biggest limitations of our current study is the limited sample size used for the performance evaluations. Due to time constraints, we could not gather the larger participant pool originally intended for this study. With a larger participant group we aimed to have three teams compete in parallel on the same structure for three rounds, where each team would either be managed by the AI PM, a human PM, or no PM for the entire study period. Given our smaller sample size we instead opted to mix the management style between nine rounds of different structures. In addition, having a larger dataset of experienced human project-managers could provide more insight into how the AI PM compares more broadly to human directed efforts. We are actively developing the codebase to accommodate for testing with a larger sample size, as such bugs experienced in the current evaluation environment are currently being resolved.

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